

Design and Simulation of a 6-Bit 5 MHz SAR ADC Using Binary Weighted Resistor DAC in 90nm CMOS Technology

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Abstract—This paper presents the design and simulation of a 6-bit Successive Approximation Register (SAR) Analog-to-Digital Converter (ADC) implemented in 90nm CMOS technology using the Cadence Virtuoso design environment. The implemented ADC architecture consists of a transmission-gate-based sample-and-hold circuit, a CMOS differential comparator, a master-slave D flip-flop based, SAR logic unit, and a binary weighted resistor digital-to-analog converter (DAC). The circuit operates from a 1 V supply voltage and supports a sampling frequency of 5 MHz, making it suitable for low-voltage mixed-signal and biomedical sensing applications. Individual building blocks were designed and verified through transient and AC simulations using the Spectre simulator. The operational amplifier used in the DAC achieves a DC gain of approximately 63 dB and a phase margin of 81°, ensuring stable operation. The complete SAR ADC consumes approximately 859 μ W of power. Simulation results verify the correct functionality of the proposed architecture and demonstrate its suitability for medium-speed, low-voltage data conversion applications.

Keywords—SAR ADC, Analog-to-Digital Converter, CMOS Comparator, Binary Weighted DAC, Cadence Virtuoso, 90nm CMOS, Low-Power Design.

I. INTRODUCTION

The rapid advancement of modern electronic systems has significantly increased the demand for efficient data conversion techniques. Most real-world signals, such as biomedical, environmental, industrial, and communication signals, exist in analog form, whereas modern processing units operate in the digital

domain. Therefore, Analog-to-Digital Converters (ADCs) play a crucial role in bridging the gap between analog signal acquisition and digital signal processing. The performance of an ADC directly affects the accuracy, power consumption, and overall efficiency of the electronic system in which it is employed.[4]

With the growing popularity of portable and battery-operated devices, power consumption has become one of the most important design considerations in mixed-signal integrated circuits. Applications such as wearable health-monitoring systems, biomedical sensors, implantable medical devices, and Internet of Things (IoT) nodes require ADC architectures that can operate with low supply voltages while maintaining acceptable conversion accuracy. [5]These applications typically process low-frequency signals and therefore do not require extremely high-speed converters. Instead, low power consumption, compact architecture, and reliable operation are the primary design objectives.[5]

Several ADC architectures have been developed over the years, including Flash ADCs, Pipeline ADCs, Sigma-Delta ADCs, Dual-Slope ADCs, and Successive Approximation Register (SAR) ADCs. Among these architectures, SAR ADCs offer an attractive balance between conversion speed, resolution, circuit complexity, and power efficiency. Unlike pipeline and flash converters, SAR ADCs require fewer analog building blocks and rely heavily on digital circuitry, making them highly suitable for implementation in scaled CMOS technologies. Their simple architecture and low power consumption have made them one of the most widely adopted ADC solutions for medium-resolution applications.[4][5]

A conventional SAR ADC consists of four primary building blocks: a Sample-and-Hold (S/H) circuit, a comparator, a Digital-to-Analog Converter (DAC), and SAR control logic. During each conversion cycle, the sampled input signal is compared with the DAC output, and the SAR logic performs a binary search algorithm to determine the digital representation of the input signal. This approach enables efficient analog-to-digital conversion with relatively low hardware complexity and power consumption.

In this work, a 6-bit SAR ADC has been designed and simulated using a 90 nm CMOS technology in the Cadence Virtuoso design environment. The proposed design operates from a 1 V supply voltage and employs a transmission-gate-based Sample-and-Hold circuit, a CMOS differential comparator, a master-slave D flip-flop based SAR logic unit, and a binary weighted resistor DAC. The ADC is designed to operate at a sampling frequency of 5 MHz while maintaining low power consumption suitable for low-voltage mixed-signal applications. The functionality of each individual block has been verified through transient and AC simulations using the Spectre simulator, followed by integration of all blocks to realize the complete SAR ADC architecture.[4][5]

II. PROPOSED SAR ADC ARCHITECTURE

The architecture of the proposed 6-bit Successive Approximation Register (SAR) Analog-to-Digital Converter is shown in Fig. 1. The converter is implemented using four major functional blocks: a Sample-and-Hold (S/H) circuit, a CMOS comparator, a Successive Approximation Register (SAR) logic unit, and a binary weighted resistor Digital-to-Analog Converter (DAC). The complete system operates from a 1 V power supply and is designed using 90 nm CMOS technology.[1]

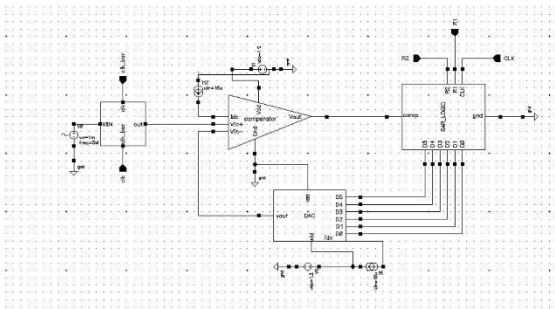


Fig. 1. Proposed 6-bit SAR ADC architecture.

At the beginning of the conversion process, the analog input signal is sampled by the Sample-and-Hold circuit. During the sampling phase, the input signal is tracked and stored on the hold capacitor. Once the sampling interval is completed, the circuit enters the hold phase, where the sampled voltage remains constant throughout the conversion process. This ensures that the input signal does not vary while the SAR algorithm is determining the corresponding digital code.[1][4][6]

The stored analog voltage is then applied to one input of the comparator, while the output voltage generated by the DAC is connected to the other comparator input. The comparator continuously compares these two voltages and produces a digital output indicating whether the sampled input voltage is greater or less than the DAC output voltage.[4]

The SAR logic unit performs the successive approximation process based on the comparator output. Initially, the Most Significant Bit (MSB) is set to logic high while all remaining bits are cleared. The resulting digital code is sent to the DAC, which generates an equivalent analog voltage.[1] The comparator evaluates the difference between the sampled input voltage and the DAC output voltage. If the DAC output exceeds the sampled input, the corresponding bit is cleared; otherwise, it is retained. The SAR logic then proceeds to the next lower bit and repeats the comparison process.[1]

This binary search operation continues until all six bits have been evaluated. Since the converter resolution is 6 bits, six comparison cycles are required to complete one conversion. After the final comparison cycle, the SAR register contains the digital representation of the sampled analog input signal.[1]

The binary weighted resistor DAC converts the SAR digital output into an equivalent analog voltage during each approximation step. The DAC output is updated dynamically as the SAR logic modifies the trial code, enabling the comparator to make the necessary decisions throughout the conversion process.[6]

The overall conversion procedure follows the principle of successive approximation, which significantly reduces hardware complexity compared to flash architectures while maintaining moderate conversion speed and low power consumption. Owing to its predominantly digital structure and reduced

analog circuitry, the proposed SAR ADC is well suited for low-voltage and medium-speed applications such as biomedical signal acquisition, sensor interfacing, and portable electronic systems.[1]

III. CIRCUIT DESIGN AND IMPLEMENTATION

The proposed 6-bit SAR ADC is composed of four major building blocks: a Sample-and-Hold (S/H) circuit, a CMOS comparator, a binary weighted resistor DAC, and a master-slave D flip-flop based SAR logic unit. Each block was individually designed and verified in Cadence Virtuoso before integrating them into the complete ADC architecture. The design objective was to achieve reliable operation at a supply voltage of 1 V while maintaining moderate conversion speed and low hardware complexity.[1][6]

A. Sample-and-Hold Circuit

The Sample-and-Hold (S/H) circuit is responsible for sampling the analog input signal and maintaining a constant voltage during the conversion process. In the proposed design, a transmission-gate-based sampling switch is employed to reduce signal degradation and improve signal tracking performance. The complementary operation of NMOS and PMOS transistors allows the transmission gate to pass both logic-high and logic-low voltage levels with minimal distortion.[1]

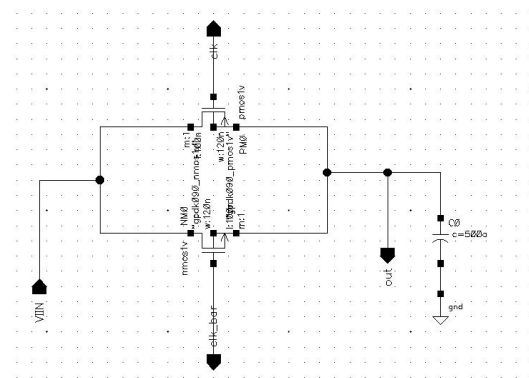


Fig. 2. Sample and Hold Circuit

During the sampling phase, the switch remains ON and the hold capacitor tracks the input signal. When the clock changes state, the switch turns OFF and the capacitor stores the sampled voltage. The stored voltage is then applied to the comparator input throughout the conversion cycle.

The transient simulation results confirm that the output follows the input signal during the sampling interval and remains stable during the hold period.

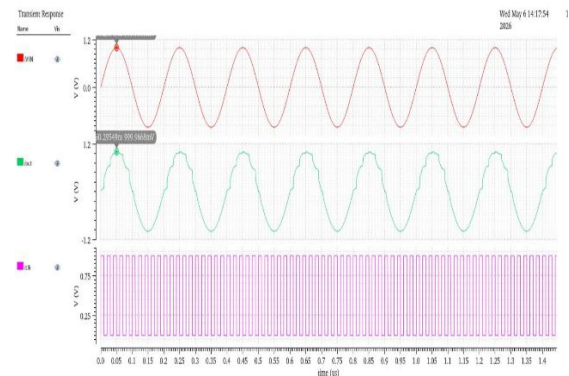


Fig. 3. S/H circuit transient analysis

B. CMOS Comparator

The comparator is one of the most critical blocks in a SAR ADC. It compares the sampled input voltage with the DAC output voltage and generates a digital decision signal for the SAR logic.

The designed comparator employs a differential input structure followed by gain and output stages. The differential configuration improves sensitivity and allows accurate comparison of small voltage differences.[1]

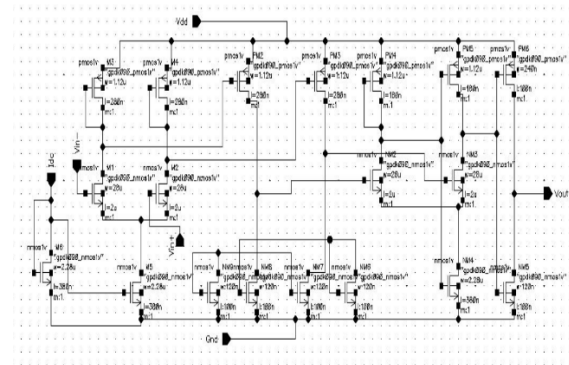


Fig. 4. CMOS Comparator Schematic

During operation, the comparator produces a logic high when the sampled input voltage is greater than the DAC output voltage. If the DAC voltage exceeds the sampled input, the output switches to logic low. This output determines whether the currently tested bit is retained or cleared in the SAR register.

Transient simulation results verify the correct switching behavior of the comparator under different input conditions.

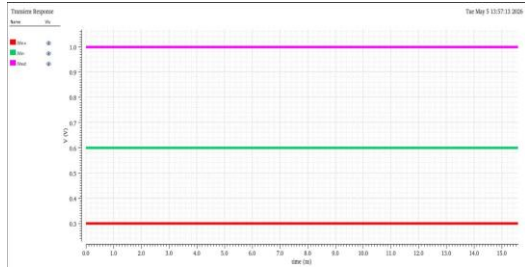


Fig. 5. Comparator Output Waveform

C. Operational Amplifier

A two-stage CMOS operational amplifier was designed and used in the DAC implementation. The first stage provides voltage gain, while the second stage improves output drive capability and overall stability.[1][3]

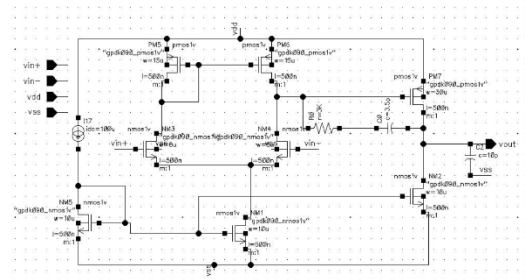


Fig. 6. Two Stage Operational Amplifier schematic

analysis was performed to evaluate the amplifier performance. The simulation results show a DC gain of approximately 63 dB, a unity gain bandwidth of about 22 MHz, and a phase margin close to 81°. These values indicate stable operation and sufficient gain for the intended application.[3]

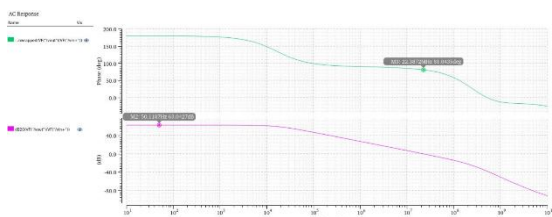


Fig. 7. Operational Amplifier Phase and Gain Margin

D. Binary Weighted Resistor DAC

The Digital-to-Analog Converter (DAC) generates the reference voltage required during each step of the successive approximation process. The accuracy of the SAR ADC largely depends on the precision of the DAC output, as every comparator decision is based on the comparison between the sampled input voltage and the DAC-generated voltage.[1][8]

In this work, a 6-bit binary weighted resistor DAC was implemented using resistor values of 1 kΩ, 2 kΩ, 4 kΩ, 8 kΩ, 16 kΩ, and 32 kΩ. The resistor network is connected to an operational amplifier configured as a summing amplifier to generate the corresponding analog output voltage.[9]

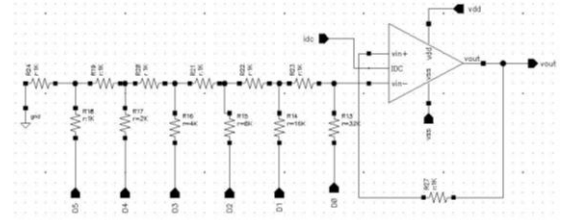
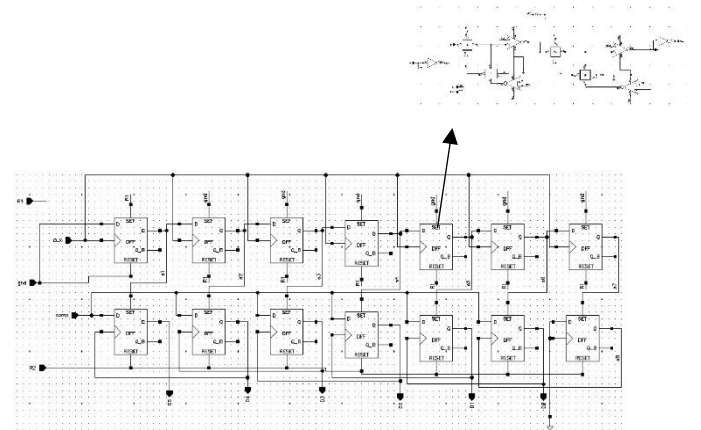


Fig.8. Binary Weighted Resistor DAC Schematic

E. SAR Logic Unit

The SAR logic unit controls the entire conversion process by implementing the successive approximation algorithm. Based on the comparator output, it determines whether each bit should remain set or be cleared, eventually generating the final digital representation of the sampled input signal.

The SAR logic was implemented using master-slave D flip-flops connected in a shift-register configuration. This approach provides reliable operation and simplifies the implementation of the binary search algorithm used in SAR ADCs.[1]



At the beginning of the conversion process, the Most Significant Bit (MSB) is set to logic high while all remaining bits are cleared. The resulting digital code is applied to the DAC, which generates an equivalent analog voltage. The comparator then determines whether the

DAC output is higher or lower than the sampled input voltage.

If the DAC output exceeds the sampled input, the currently tested bit is cleared. Otherwise, the bit remains set. The SAR logic then proceeds to the next lower bit and repeats the process until all six bits have been evaluated. The operation of the master-slave D flip-flop was verified through transient simulation. The results confirm correct data storage and transfer at the clock transitions.

The complete SAR logic timing sequence was also simulated to verify the correct execution of the successive approximation algorithm. The waveform demonstrates the sequential evaluation of each bit during the conversion cycle.

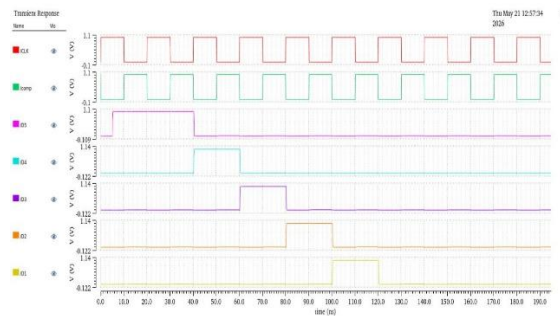


Fig. 12. SAR Logic Timing Waveform

IV. COMPLETE SAR ADC PERFORMANCE AND DISCUSSION

After verifying the functionality of the individual circuit blocks, the Sample-and-Hold circuit, comparator, DAC, and SAR logic were integrated to realize the complete 6-bit SAR ADC architecture. The integrated design was simulated in the Cadence Virtuoso environment using the Spectre simulator to evaluate the overall performance of the converter.

The proposed ADC operates from a 1 V supply voltage and supports a sampling frequency of 5 MHz. The SAR logic performs a successive approximation routine over six clock cycles, allowing the analog input signal to be converted into its corresponding 6-bit digital representation.

The overall design consumes approximately 859 μW of power under nominal operating conditions. The use of a low supply voltage and a predominantly digital architecture contributes to reduced power consumption while maintaining reliable operation.

Parameter	Value
ADC Architecture	SAR ADC
Resolution	6-bit
Technology	90 nm CMOS
Supply Voltage	1 V
Sampling Frequency	5 MHz
DAC Type	Binary Weighted Resistor DAC
SAR Logic	Master-Slave D Flip-Flop Based
Power Consumption	859 μW
Simulator	Cadence Spectre

Table I. Performance Summary of the Proposed SAR ADC

To evaluate the implemented design, a comparison with previously reported SAR ADCs is presented in Table II. Although the proposed design operates at a higher sampling frequency than some reported implementations, it exhibits higher power consumption due to the selected circuit architecture and operating conditions.

Reference	Kulkarni et al.	Pham et al.	Proposed work
Technology	90nm	65nm	90nm
Resolution	6-bit	6-bit	6-bit
Supply Voltage	1.2V	1.0V	1.0V
Sampling Frequency	—	2kHz	5Mhz
Power Consumption	802.6 μW	883.7 μW	859 μW

Table II. Performance Comparison with Existing Works

The simulation results confirm the successful integration of all building blocks and demonstrate the feasibility of implementing a low-voltage 6-bit SAR ADC using standard 90 nm CMOS technology. The converter achieves stable operation at 5 MHz while maintaining moderate power consumption, making it suitable for low-power mixed-signal applications.

V. CONCLUSION

This paper presented the design and simulation of a 6-bit Successive Approximation Register (SAR) Analog-to-Digital Converter implemented in a 90 nm CMOS process using the Cadence Virtuoso design environment. The proposed ADC consists of a transmission-gate-

based Sample-and-Hold circuit, a CMOS comparator, a binary weighted resistor DAC, and a master-slave D flip-flop based SAR logic unit.

Each functional block was designed and verified individually before integrating them into the complete ADC architecture. The operational amplifier achieved a DC gain of approximately 63 dB with a phase margin of 81°, ensuring stable operation within the DAC structure. The complete ADC operates from a 1 V supply voltage at a sampling frequency of 5 MHz and consumes approximately 859 μ W of power.

The simulation results demonstrate the successful implementation of the SAR conversion process and validate the functionality of the proposed design. Owing to its simple architecture, low supply voltage, and moderate operating speed, the developed ADC can be considered suitable for low-power mixed-signal applications, sensor interfaces, and biomedical signal acquisition systems.

Future work may include layout implementation, post-layout verification, optimization of power consumption, and the evaluation of additional performance parameters such as ENOB, SNDR, DNL, and INL.

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